Status	Finished
Started	Monday, 5 October 2026, 11:01 AM
Completed	Monday, 5 October 2026, 11:05 AM
Duration	4 mins 21 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Complete

Mark 1.00 out of 1.00

What is a significant characteristic of Bubble Sort?

- a. It bubbles up the largest element in each pass
- b. It recursively sorts subproblems
- c. It divides the list into sublists
- d. It uses the heap data structure

The correct answer is: It bubbles up the largest element in each pass

Question **2**

Complete

Mark 1.00 out of 1.00

What is one advantage of sorting a list before performing a search operation?

- a. It makes the search operation slower
- b. It allows for faster searching
- c. It has no effect on the search operation
- d. It increases the number of comparisons needed

The correct answer is: It allows for faster searching

Question **3**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

- a. 200,470,80,150,40,90,300,400,70,120
- b. 40,80,90,150,200,300,400,470,70,120
- c. 80,150,200,470,40,90,300,400,70,120
- d. 40,70,80,90,120,150,200,300,400,470

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **4**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

- a. It avoids the need for recursion
- b. It simplifies the sorting process by using only one pass
- c. It allows for efficient parallel processing and sorting of data
- d. It only works on small datasets

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **5**

Complete

Mark 1.00 out of 1.00

What is a key disadvantage of Bubble Sort compared to more advanced algorithms like Merge Sort?

- a. Bubble Sort cannot handle duplicate elements
- b. Bubble Sort is difficult to implement
- c. Bubble Sort does not guarantee sorted order
- d. Bubble Sort is less efficient for large lists

The correct answer is: Bubble Sort is less efficient for large lists

Question **6**

Complete

Mark 1.00 out of 1.00

Which algorithm typically follows a divide-and-conquer structure?

a. Linear Search
b. Merge Sort
c. Binary Search
d. Bubble Sort

The correct answer is: Merge Sort

Question **7**

Complete

Mark 1.00 out of 1.00

What is sorting in the context of computer science?

a. Inserting data into a list
b. Arranging data in a particular format
c. Searching for data in a list
d. Deleting data from a list

The correct answer is: Arranging data in a particular format

Question **8**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is known for its simplicity and sequential comparison of elements?

a. Merge Sort
b. Heap Sort
c. Quick Sort
d. Bubble Sort

The correct answer is: Bubble Sort

Question **9**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is described as making multiple passes through a list, comparing elements, and swapping adjacent items that are out of order?

a. Merge Sort
b. Insertion Sort
c. Bubble Sort
d. Quick Sort

The correct answer is: Bubble Sort

Question **10**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

a. Bubble Sort
b. Linear Search
c. Merge Sort
d. Binary Search

The correct answer is: Bubble Sort

Question **11**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

a. They are split again into smaller sublists
b. They are discarded
c. They are combined to form a single sorted list
d. They are compared element by element

The correct answer is: They are combined to form a single sorted list

Question **12**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is based on repeatedly dividing the list into halves?

a. Bubble Sort
b. Quick Sort
c. Merge Sort
d. Insertion Sort

The correct answer is: Merge Sort

Question **13**

Complete

Mark 1.00 out of 1.00

The process of placing or rearranging a collection of elements into a particular order is known as

a. Searching
b. Sorting
c. Merging
d. Rearranging

The correct answer is: Sorting

Question **14**

Complete

Mark 0.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

a. Quick Sort
b. Bubble Sort
c. Linear Search
d. Merge Sort

The correct answer is: Merge Sort

Question **15**

Complete

Mark 1.00 out of 1.00

Which Python function would you use to sort a list in-place?

- a. sort()
- b. arrange()
- c. order()
- d. sorted()

The correct answer is: sort()